```
SIMILAR_SEGMENTS_SQL = """
SELECT p.title, s.id, s.content, s.start_time, s.end_time,
      s.embedding <-> source.embedding AS distance
FROM segment AS s
JOIN podcast AS p ON p.id = s.podcast_id
CROSS JOIN (SELECT embedding FROM segment WHERE id = %s) AS source
WHERE s.id <> %s
ORDER BY distance ASC, s.id ASC
LIMIT %s;
"""

DISTANT_SEGMENTS_SQL = """
SELECT p.title, s.id, s.content, s.start_time, s.end_time,
      s.embedding <-> source.embedding AS distance
FROM segment AS s
JOIN podcast AS p ON p.id = s.podcast_id
CROSS JOIN (SELECT embedding FROM segment WHERE id = %s) AS source
WHERE s.id <> %s
ORDER BY distance DESC, s.id ASC
LIMIT %s;
"""

SIMILAR_EPISODES_TO_SEGMENT_SQL = """
WITH episode_embeddings AS (
   SELECT podcast_id, AVG(embedding) AS embedding
   FROM segment
   GROUP BY podcast_id
), source AS (
   SELECT podcast_id, embedding
   FROM segment
   WHERE id = %s
)
SELECT p.title,
      episode_embeddings.embedding <-> source.embedding AS distance
FROM episode_embeddings
JOIN podcast AS p ON p.id = episode_embeddings.podcast_id
CROSS JOIN source
WHERE episode_embeddings.podcast_id <> source.podcast_id
ORDER BY distance ASC, p.id ASC
LIMIT %s;
"""

SIMILAR_EPISODES_SQL = """
WITH episode_embeddings AS (
```

```
    SELECT podcast_id, AVG(embedding) AS embedding
    FROM segment
    GROUP BY podcast_id
), source AS (
    SELECT embedding
    FROM episode_embeddings
    WHERE podcast_id = %s
)
SELECT p.title,
       episode_embeddings.embedding <-> source.embedding AS distance
FROM episode_embeddings
JOIN podcast AS p ON p.id = episode_embeddings.podcast_id
CROSS JOIN source
WHERE episode_embeddings.podcast_id <> %s
ORDER BY distance ASC, p.id ASC
LIMIT %s;
"""


Question 1: five most similar segments to 267:476
  title='Podcast: Ryan Graves: UFOs, Fighter Jets, and Aliens | Lex Fridman Podcast #308' |
segment=113:2792 | start=6725.62 | stop=6729.86 | distance=0.648345
     encounters, human beings, if we were to meet another alien
  title='Podcast: Richard Dawkins: Evolution, Intelligence, Simulation, and Memes | Lex Fridman
Podcast #87' | segment=268:1019 | start=2900.04 | stop=2903.0800000000004 |
distance=0.655811
     Suppose we did meet an alien from outer space
  title='Podcast: Jeffrey Shainline: Neuromorphic Computing and Optoelectronic Intelligence |
Lex Fridman Podcast #225' | segment=305:3600 | start=9479.960000000001 | stop=9484.04 |
distance=0.659543
     but if we think of alien civilizations out there
  title='Podcast: Michio Kaku: Future of Humans, Aliens, Space Travel & Physics | Lex Fridman
Podcast #45' | segment=18:464 | start=1316.8600000000001 | stop=1319.5800000000002 |
distance=0.666203
     So I think when we meet alien life from outer space,
  title='Podcast: Alien Debate: Sara Walker and Lee Cronin | Lex Fridman Podcast #279' |
segment=71:989 | start=2342.34 | stop=2343.6200000000003 | distance=0.674294
     because if aliens come to us

Question 2: five most dissimilar segments to 267:476
  title='Podcast: Jason Calacanis: Startups, Angel Investing, Capitalism, and Friendship | Lex
Fridman Podcast #161' | segment=119:218 | start=519.96 | stop=523.8000000000001 |
distance=1.615769
     a 73 Mustang Grande in gold?
```

title='Podcast: Rana el Kaliouby: Emotion AI, Social Robots, and Self-Driving Cars | Lex Fridman Podcast #322' | segment=133:2006 | start=5818.62 | stop=5820.82 | distance=1.586336
for 94 car models.
title='Podcast: Travis Stevens: Judo, Olympics, and Mental Toughness | Lex Fridman Podcast #223' | segment=283:1488 | start=3709.34 | stop=3711.1000000000004 | distance=1.572553
when I called down to get the sauna.
title='Podcast: Jeremy Howard: fast.ai Deep Learning Courses and Research | Lex Fridman Podcast #35' | segment=241:1436 | start=4068.9 | stop=4071.1400000000003 | distance=1.566332
which has all the courses pre-installed.
title='Podcast: Joscha Bach: Nature of Reality, Dreams, and Consciousness | Lex Fridman Podcast #212' | segment=307:3933 | start=10648.64 | stop=10650.960000000001 | distance=1.561634
and very few are first class and some are budget.

Question 3: five most similar segments to 48:511
title='Podcast: Andrew Huberman: Neuroscience of Optimal Performance | Lex Fridman Podcast #139' | segment=155:648 | start=3798.48 | stop=3805.84 | distance=0.652300
Is there something interesting to you or fundamental to you about the circuitry of the brain
title='Podcast: Cal Newport: Deep Work, Focus, Productivity, Email, and Social Media | Lex Fridman Podcast #166' | segment=61:3707 | start=8498.02 | stop=8500.1 | distance=0.712105
of what we might discover about neural networks?
title='Podcast: Matt Botvinick: Neuroscience, Psychology, and AI at DeepMind | Lex Fridman Podcast #106' | segment=48:512 | start=1846.84 | stop=1865.84 | distance=0.719560
And our brain is there. There's some there's quite a few differences. Are some of them to you either interesting or perhaps profound in terms of in terms of the gap we might want to try to close in trying to create a human level intelligence.
title='Podcast: Yann LeCun: Dark Matter of Intelligence and Self-Supervised Learning | Lex Fridman Podcast #258' | segment=276:2642 | start=8628.16 | stop=8646.16 | distance=0.735722
Have these, I mean, small pockets of beautiful complexity. Does that, do cellular automata, do these kinds of emergence and complex systems give you some intuition or guide your understanding of machine learning systems and neural networks and so on?
title='Podcast: Stephen Wolfram: Fundamental Theory of Physics, Life, and the Universe | Lex Fridman Podcast #124' | segment=2:152 | start=610.86 | stop=618.86 | distance=0.736697
So is there something like that with physics where so deep learning neural networks have been around for a long time?

Question 4: five most similar segments to 51:56
title='Podcast: George Hotz: Hacking the Simulation & Learning to Drive with Neural Nets | Lex Fridman Podcast #132' | segment=308:144 | start=500.44 | stop=502.6 | distance=0.668197
I mean, we don't understand dark energy, right?

title='Podcast: Lex Fridman: Ask Me Anything - AMA January 2021 | Lex Fridman Podcast' | segment=243:273 | start=946.22 | stop=950.12 | distance=0.735551
Like, what's up with this dark matter and dark energy stuff?
title='Podcast: Katherine de Kleer: Planets, Moons, Asteroids & Life in Our Solar System | Lex Fridman Podcast #184' | segment=196:685 | start=2591.72 | stop=2595.9599999999996 | distance=0.763114
being like, what the hell is dark matter and dark energy?
title='Podcast: Alex Filippenko: Supernovae, Dark Energy, Aliens & the Expanding Universe | Lex Fridman Podcast #137' | segment=51:36 | start=216.0 | stop=219.0 | distance=0.792202
Do we have any understanding of what the heck that thing is?
title='Podcast: Leonard Susskind: Quantum Mechanics, String Theory and Black Holes | Lex Fridman Podcast #41' | segment=122:831 | start=2374.9 | stop=2377.6200000000003 | distance=0.802270
That is a big question in physics right now.

Question 5: five episodes most similar to segment 267:476
title='Podcast: Sara Walker: The Origin of Life on Earth and Alien Worlds | Lex Fridman Podcast #198' | distance=0.782898
title='Podcast: Martin Rees: Black Holes, Alien Life, Dark Matter, and the Big Bang | Lex Fridman Podcast #305' | distance=0.787950
title='Podcast: Max Tegmark: Life 3.0 | Lex Fridman Podcast #1' | distance=0.788690
title='Podcast: Sean Carroll: The Nature of the Universe, Life, and Intelligence | Lex Fridman Podcast #26' | distance=0.789065
title='Podcast: Nick Bostrom: Simulation and Superintelligence | Lex Fridman Podcast #83' | distance=0.791121

Question 5: five episodes most similar to segment 48:511
title='Podcast: Christof Koch: Consciousness | Lex Fridman Podcast #2' | distance=0.753780
title='Podcast: Dileep George: Brain-Inspired AI | Lex Fridman Podcast #115' | distance=0.760515
title='Podcast: Tomaso Poggio: Brains, Minds, and Machines | Lex Fridman Podcast #13' | distance=0.761555
title='Podcast: Elon Musk: Neuralink, AI, Autopilot, and the Pale Blue Dot | Lex Fridman Podcast #49' | distance=0.776152
title='Podcast: Philip Goff: Consciousness, Panpsychism, and the Philosophy of Mind | Lex Fridman Podcast #261' | distance=0.787206

Question 5: five episodes most similar to segment 51:56
title='Podcast: Sean Carroll: Quantum Mechanics and the Many-Worlds Interpretation | Lex Fridman Podcast #47' | distance=0.776714
title='Podcast: Stephen Wolfram: Fundamental Theory of Physics, Life, and the Universe | Lex Fridman Podcast #124' | distance=0.808071
title='Podcast: Donald Hoffman: Reality is an Illusion - How Evolution Hid the Truth | Lex Fridman Podcast #293' | distance=0.816583

  title='Podcast: Cumrun Vafa: String Theory | Lex Fridman Podcast #204' | distance=0.817347
  title='Podcast: Avi Loeb: Aliens, Black Holes, and the Mystery of the Oumuamua | Lex Fridman Podcast #154' | distance=0.825452

Question 6: five episodes most similar to episode VeH7qKZr0WI
  title='Podcast: Tyler Cowen: Economic Growth & the Fight Against Conformity & Mediocrity | Lex Fridman Podcast #174' | distance=0.119501
  title='Podcast: Eric Weinstein: Difficult Conversations, Freedom of Speech, and Physics | Lex Fridman Podcast #163' | distance=0.125714
  title='Podcast: Michael Malice and Yaron Brook: Ayn Rand, Human Nature, and Anarchy | Lex Fridman Podcast #178' | distance=0.128427
  title='Podcast: Steve Keen: Marxism, Capitalism, and Economics | Lex Fridman Podcast #303' | distance=0.129163
  title='Podcast: Michael Malice: The White Pill, Freedom, Hope, and Happiness Amidst Chaos | Lex Fridman Podcast #150' | distance=0.130409